

Homework #6 - Spaceflight Mechanics

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1 Part 1

1.1 Problem

Chosen an initial position of the body with respect to the reference F_N (in particular by assigning the direction of $\hat{b}_1$ and the meridian on which $\hat{b}_2$ lies) via the *EXCEL* sheet.

- locate $\hat{e}$ and Φ
- rotate the body increasing Φ by 10° (in 1 s)
- rotate the body again increasing φ by 20° (in 2 s)
- rotate the body again increasing θ by 30° (in 3 s)
- rotate the body again increasing ψ by 40° (in 4 s)

Take note of the body positions as described by the different sets of variables.

1.2 Body positions

The many positions the body occupies are written in 5 different ways each. Also the body kinematics over a span of around 13 s are shown as if the body were starting in the considered position. These are obtained through integration, as shown in the Appendix.

- Position 1 : initial attitude
 $\alpha_1 = 40^\circ, \delta_1 = 20^\circ, \alpha_2 = -15^\circ$

$$BN = \begin{bmatrix} 0.7198 & 0.6040 & 0.3420 \\ 0.5175 & -0.1387 & -0.8843 \\ -0.4626 & 0.7848 & -0.4124 \end{bmatrix}$$

$$\begin{cases} [\alpha_1, \delta_1] = [40^\circ, 20^\circ] \\ [\alpha_2, \delta_2] = [-15^\circ, -57.60^\circ] \\ [\alpha_3, \delta_3] = [120.52^\circ, -24.36^\circ] \end{cases}$$

$$\begin{cases} [\alpha_E, \delta_E] = [-153.72^\circ, 2.73^\circ] \\ \Phi = 114.56^\circ \end{cases}$$

$$\begin{cases} \varphi = -116.03^\circ \\ \theta = -20^\circ \\ \psi = 40^\circ \end{cases}$$

$$\begin{cases} q_0 = 0.5405 \\ q_1 = -0.7535 \\ q_2 = -0.3721 \\ q_3 = 0.0400 \end{cases}$$

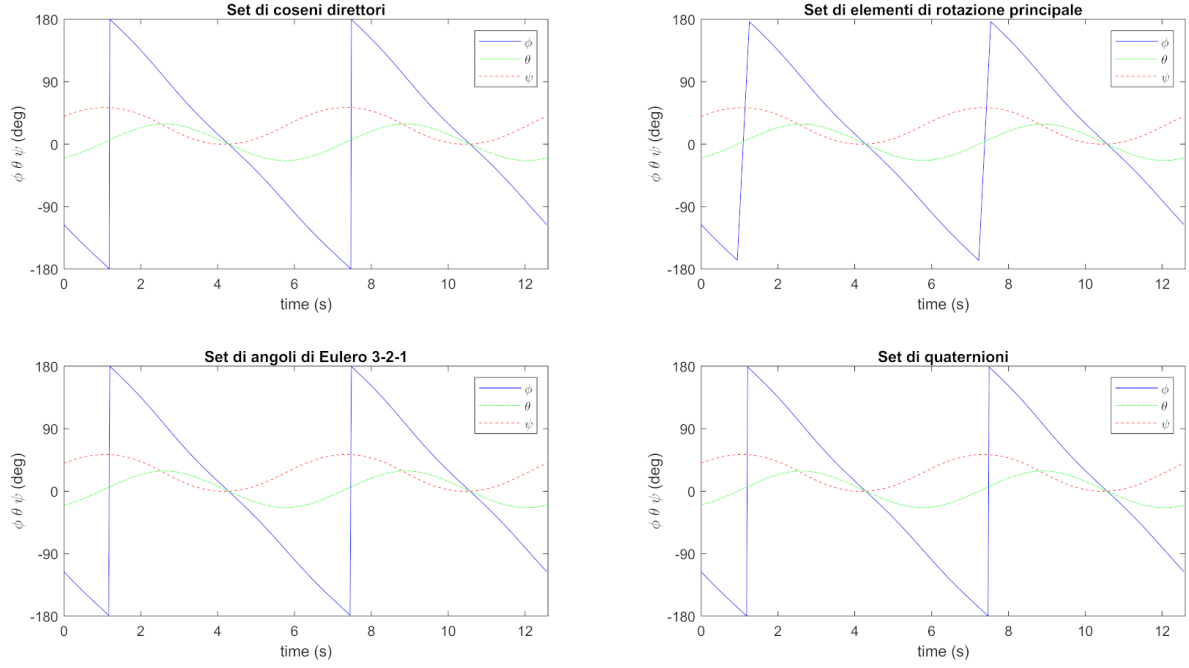


FIGURE 1. Body kinematics starting from position 1

- Position 2 : after rotation of Φ by 10° (in 1 s)
 $\alpha_E = -153.72^\circ$, $\delta_E = 2.73^\circ$, $\Phi = 114.56^\circ + 10^\circ = 124.56^\circ$

$$BN = \begin{bmatrix} 0.6898 & 0.6600 & 0.2975 \\ 0.5817 & -0.2606 & -0.7705 \\ -0.4310 & 0.7046 & -0.5637 \end{bmatrix}$$

$$\begin{cases} [\alpha_1, \delta_1] = [43.73^\circ, 17.31^\circ] \\ [\alpha_2, \delta_2] = [-24.14^\circ, -50.40^\circ] \\ [\alpha_3, \delta_3] = [121.45^\circ, -34.31^\circ] \end{cases}$$

$$\begin{cases} [\alpha_E, \delta_E] = [-153.72^\circ, 2.73^\circ] \\ \Phi = 124.56^\circ \end{cases}$$

$$\begin{cases} \varphi = -126.19^\circ \\ \theta = -17.31^\circ \\ \psi = 43.73^\circ \end{cases}$$

$$\begin{cases} q_0 = 0.4652 \\ q_1 = -0.7000 \\ q_2 = -0.3783 \\ q_3 = 0.1094 \end{cases}$$

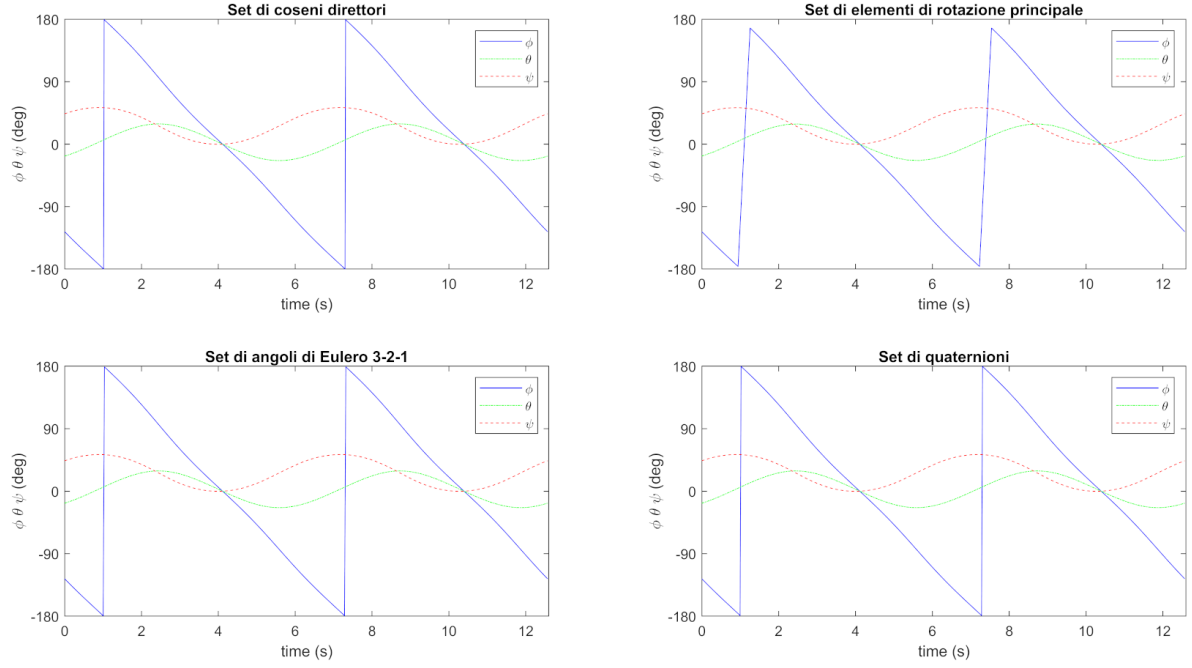


FIGURE 2. Body kinematics starting from position 2

- Position 3 : after rotation of φ by 20° (in 2 s)
 $\varphi = -126.19.72^\circ + 20^\circ = -106.19^\circ$, $\theta = -17.31^\circ$, $\psi = 43.73^\circ$

$$BN = \begin{bmatrix} 0.6898 & 0.6600 & 0.2975 \\ 0.3992 & -0.0039 & -0.9169 \\ -0.6040 & 0.7513 & -0.2662 \end{bmatrix}$$

$$\begin{cases} [\alpha_1, \delta_1] = [43.73^\circ, 17.31^\circ] \\ [\alpha_2, \delta_2] = [-0.56^\circ, -66.47^\circ] \\ [\alpha_3, \delta_3] = [128.80^\circ, -15.44^\circ] \end{cases}$$

$$\begin{cases} [\alpha_E, \delta_E] = [-151.61^\circ, 7.83^\circ] \\ \Phi = 106.87^\circ \end{cases}$$

$$\begin{cases} \varphi = -106.19^\circ \\ \theta = -17.31^\circ \\ \psi = 43.73^\circ \end{cases}$$

$$\begin{cases} q_0 = 0.5958 \\ q_1 = -0.7000 \\ q_2 = -0.3783 \\ q_3 = 0.1094 \end{cases}$$

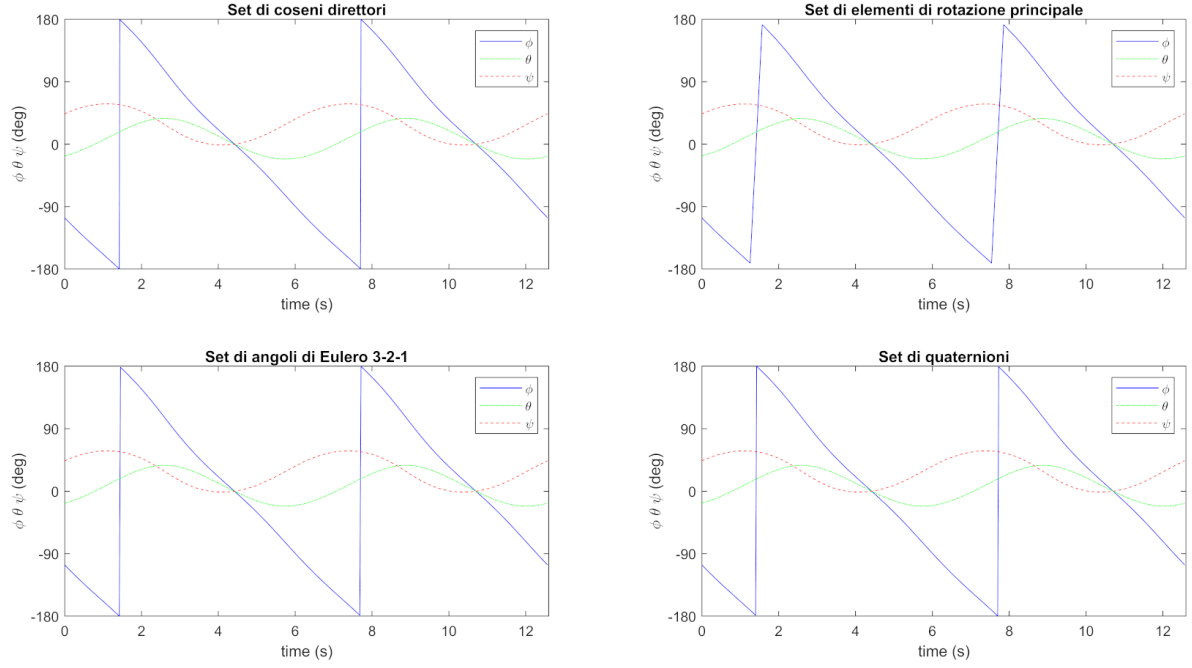


FIGURE 3. Body kinematics starting from position 3

- Position 4 : after rotation of θ by 30° (in 3 s)
 $\varphi = -106.19^\circ$, $\theta = -17.31^\circ + 30^\circ = 12.69^\circ$, $\psi = 43.73^\circ$

$$BN = \begin{bmatrix} 0.7049 & 0.6744 & -0.2197 \\ 0.0403 & -0.3473 & -0.9369 \\ -0.7082 & 0.6516 & -0.2720 \end{bmatrix}$$

$$\begin{cases} [\alpha_1, \delta_1] = [43.73^\circ, -12.69^\circ] \\ [\alpha_2, \delta_2] = [-83.38^\circ, -69.54^\circ] \\ [\alpha_3, \delta_3] = [137.38^\circ, -15.78^\circ] \end{cases}$$

$$\begin{cases} [\alpha_E, \delta_E] = [-162.91^\circ, 20.89^\circ] \\ \Phi = 117.21^\circ \end{cases}$$

$$\begin{cases} \varphi = -106.19^\circ \\ \theta = 12.69^\circ \\ \psi = 43.73^\circ \end{cases}$$

$$\begin{cases} q_0 = 0.5210 \\ q_1 = -0.7623 \\ q_2 = -0.2344 \\ q_3 = 0.3043 \end{cases}$$

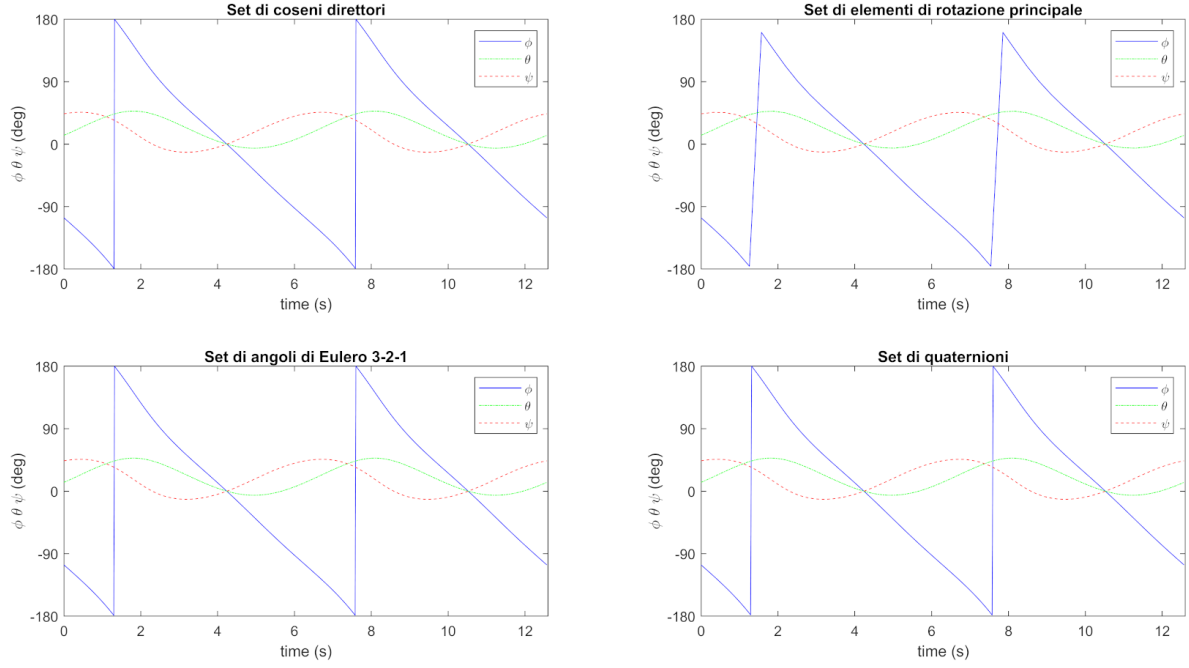


FIGURE 4. Body kinematics starting from position 4

- Position 5 : after rotation of ψ by 40° (in 4 s)
 $\varphi = -106.19^\circ$, $\theta = 12.69^\circ$, $\psi = 43.73^\circ + 40^\circ = 83.73^\circ$

$$BN = \begin{bmatrix} 0.1065 & 0.9697 & -0.2197 \\ 0.2541 & -0.2401 & -0.9369 \\ -0.9613 & 0.0439 & -0.2720 \end{bmatrix}$$

$$\begin{cases} [\alpha_1, \delta_1] = [83.73^\circ, -12.69^\circ] \\ [\alpha_2, \delta_2] = [-43.38^\circ, -69.54^\circ] \\ [\alpha_3, \delta_3] = [177.38^\circ, -15.78^\circ] \end{cases}$$

$$\begin{cases} [\alpha_E, \delta_E] = [-142.91^\circ, 30.20^\circ] \\ \Phi = 134.65^\circ \end{cases}$$

$$\begin{cases} \varphi = -106.19^\circ \\ \theta = 12.69^\circ \\ \psi = 83.73^\circ \end{cases}$$

$$\begin{cases} q_0 = 0.3855 \\ q_1 = -0.6361 \\ q_2 = -0.4810 \\ q_3 = 0.4641 \end{cases}$$

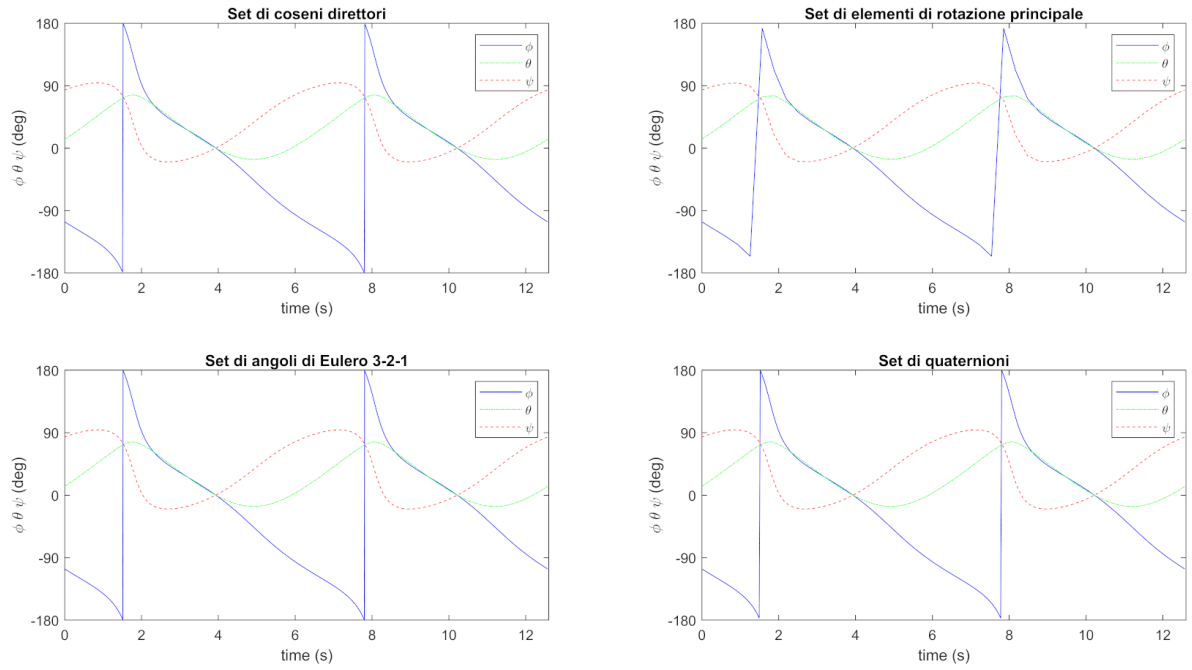


FIGURE 5. Body kinematics starting from position 5

2 Part 2

2.1 Problem

Calculate the corresponding rotational speeds in the body reference.

2.2 Estimation of the rotational speeds

We need to transform back the rotational speed of the body in the body reference.

- 1 $\rightarrow$ 2)

$$\dot{\Phi} = 10^\circ/s$$

$$\begin{cases} \alpha_{E1} = -15.72^\circ \\ \delta_{E1} = 2.73^\circ \end{cases} \Rightarrow \vec{E}_1 = [\cos \alpha_{E1} \cos \delta_{E1} \quad \sin \alpha_{E1} \cos \delta_{E1} \quad \sin \delta_{E1}]$$

$$\vec{\omega}_{12} = \dot{\Phi} \vec{E}_1 = [-8.9560^\circ \quad -4.4231^\circ \quad 0.7454^\circ]$$

$${}^B \underline{\omega} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \varphi & \sin \varphi \cos \theta \\ 0 & -\sin \varphi & \cos \varphi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

- 2 $\rightarrow$ 3)

$$\dot{\varphi} = 10^\circ/s, \dot{\theta} = 0^\circ/s, \dot{\psi} = 0^\circ/s$$

$$\vec{\omega}_{23} = \dot{\varphi} [1 \quad 0 \quad 0] = [10^\circ \quad 0^\circ \quad 0^\circ]$$

- 3 $\rightarrow$ 4)

$$\dot{\varphi} = 0^\circ/s, \dot{\theta} = 10^\circ/s, \dot{\psi} = 0^\circ/s$$

$$\vec{\omega}_{34} = \dot{\theta} [0 \quad \cos \varphi_3 \quad -\sin \varphi_3] = [0^\circ \quad -2.7880^\circ \quad 9.6035^\circ]$$

- 4 $\rightarrow$ 5)

$$\dot{\varphi} = 0^\circ/s, \dot{\theta} = 0^\circ/s, \dot{\psi} = 10^\circ/s$$

$$\vec{\omega}_{45} = \dot{\psi} [-\sin \theta_4 \quad \sin \varphi_4 \cos \theta_4 \quad \cos \varphi_4 \cos \theta_4] = [-2.1969^\circ \quad -9.3689^\circ \quad -2.7199^\circ]$$

3 Part 3

3.1 Problem

Integrate the kinematic equations using quaternions and another set of variables of your choice, verifying that, after each rotation, the body has reached the expected position.

3.2 Solution

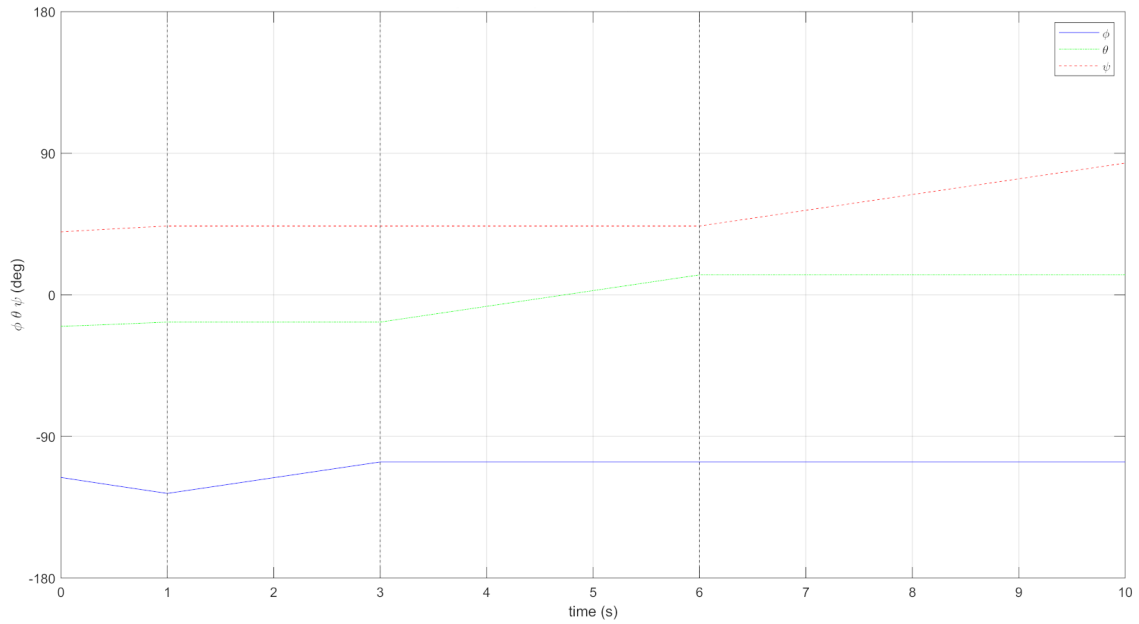


FIGURE 6. Trend of φ , θ and ψ using quaternions integration

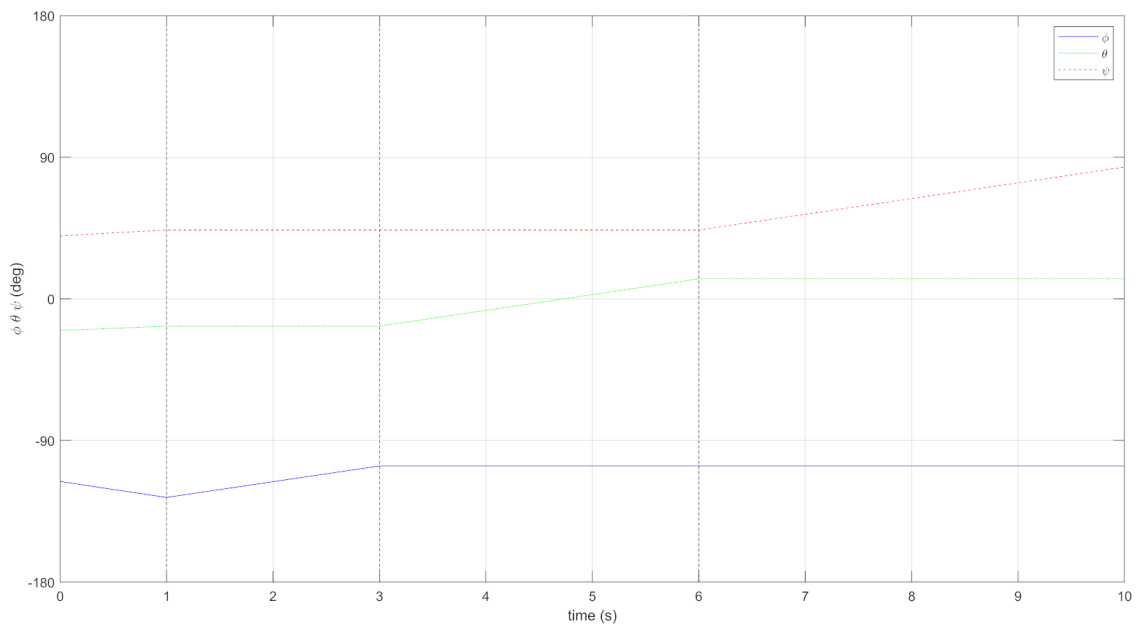


FIGURE 7. Trend of φ , θ and ψ using Euler angles (3-2-1) integration

4 Part 4

4.1 Problem

Repeat the computations with rotations ten times larger.

4.2 Solution

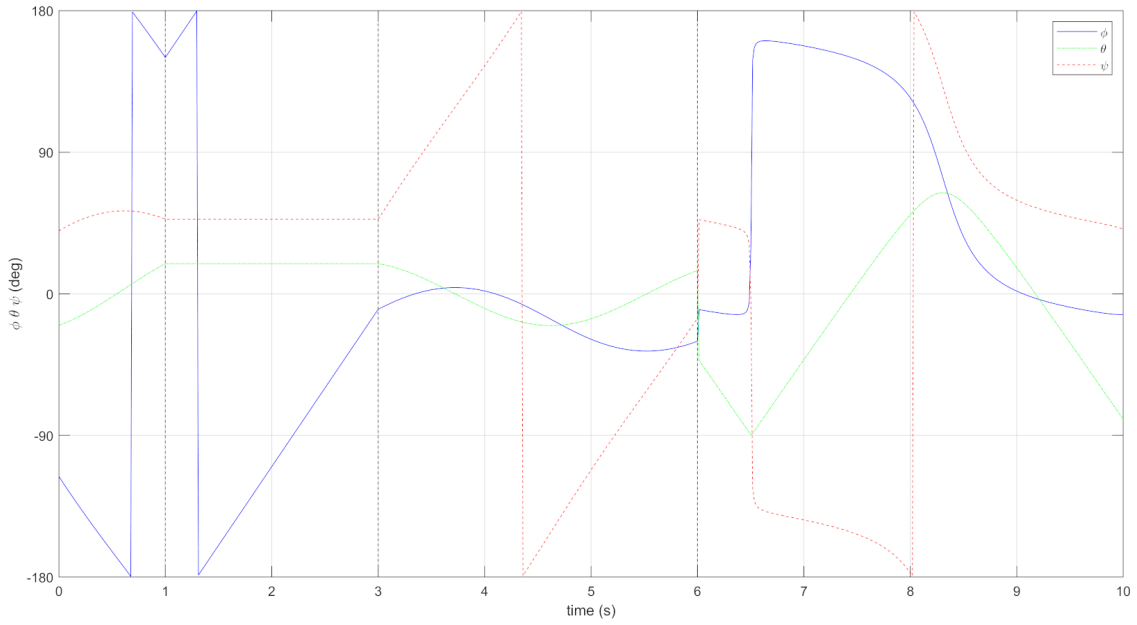


FIGURE 8. Trend of φ , θ and ψ using quaternions integration

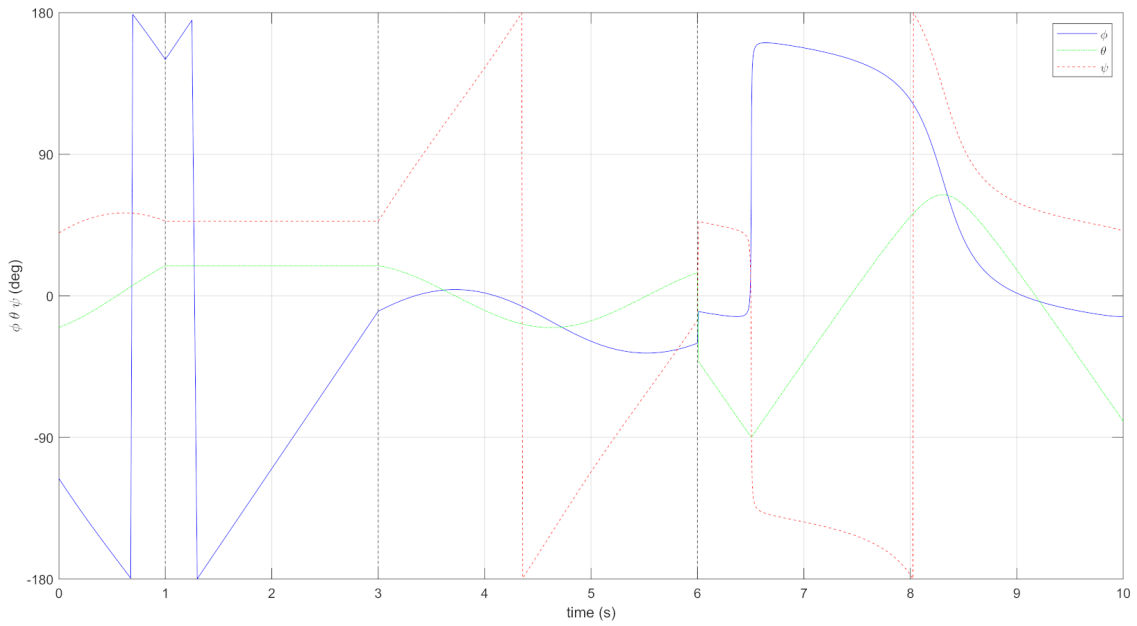


FIGURE 9. Trend of φ , θ and ψ using Euler angles (3-2-1) integration

5 Appendix

5.1 Direction cosines integration

$$\dot{BN} = -[\vec{\omega}] \cdot BN$$

$$\dot{c}_{11} = \omega_3 c_{21} - \omega_2 c_{31}$$

$$\dot{c}_{12} = \omega_3 c_{22} - \omega_2 c_{32}$$

$$\dot{c}_{21} = \omega_1 c_{31} - \omega_3 c_{11}$$

$$\dot{c}_{22} = \omega_1 c_{32} - \omega_3 c_{12}$$

$$\dot{c}_{31} = \omega_2 c_{11} - \omega_1 c_{21}$$

$$\dot{c}_{32} = \omega_2 c_{12} - \omega_1 c_{22}$$

5.2 Primary rotation integration

$$\dot{\Phi} = \hat{e} \cdot \vec{\omega}$$

$$\dot{e} = \frac{1}{2} \left[\tilde{e} - \cot \left(\frac{\Phi}{2} \right) \tilde{e} \tilde{e} \right]$$

$$\dot{\Phi} = e_1 \omega_1 + e_2 \omega_2 + e_3 \omega_3$$

$$\dot{e}_1 = \frac{1}{2} \left(e_2 \omega_3 - e_3 \omega_2 + \cot \left(\frac{\Phi}{2} \right) \cdot \omega_1 (e_2^2 + e_3^2) - e_1 e_2 \omega_2 - e_1 e_3 \omega_3 \right)$$

$$\dot{e}_2 = \frac{1}{2} \left(e_3 \omega_1 - e_1 \omega_3 + \cot \left(\frac{\Phi}{2} \right) \cdot \omega_2 (e_1^2 + e_3^2) - e_1 e_2 \omega_1 - e_2 e_3 \omega_3 \right)$$

$$\dot{e}_3 = \frac{1}{2} \left(e_1 \omega_2 - e_2 \omega_1 + \cot \left(\frac{\Phi}{2} \right) \cdot \omega_3 (e_2^2 + e_1^2) - e_1 e_3 \omega_1 - e_2 e_3 \omega_2 \right)$$

5.3 Euler angles integration

$$\begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \frac{1}{\cos \theta} \begin{bmatrix} \cos \theta & \sin \varphi \sin \theta & \cos \varphi \sin \theta \\ 0 & \cos \varphi \cos \theta & -\sin \varphi \cos \theta \\ 0 & \sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

5.4 Quaternions integration

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -\omega_1 & -\omega_2 & -\omega_3 \\ \omega_1 & 0 & \omega_3 & -\omega_2 \\ \omega_2 & -\omega_3 & 0 & \omega_1 \\ \omega_3 & \omega_2 & -\omega_1 & 0 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

5.5 Skew matrix

$$[\tilde{a}] = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$